

Exploration vs. Fixation: Scaffolding Divergent and Convergent Thinking for Human–AI Co - Creation with Generative Models

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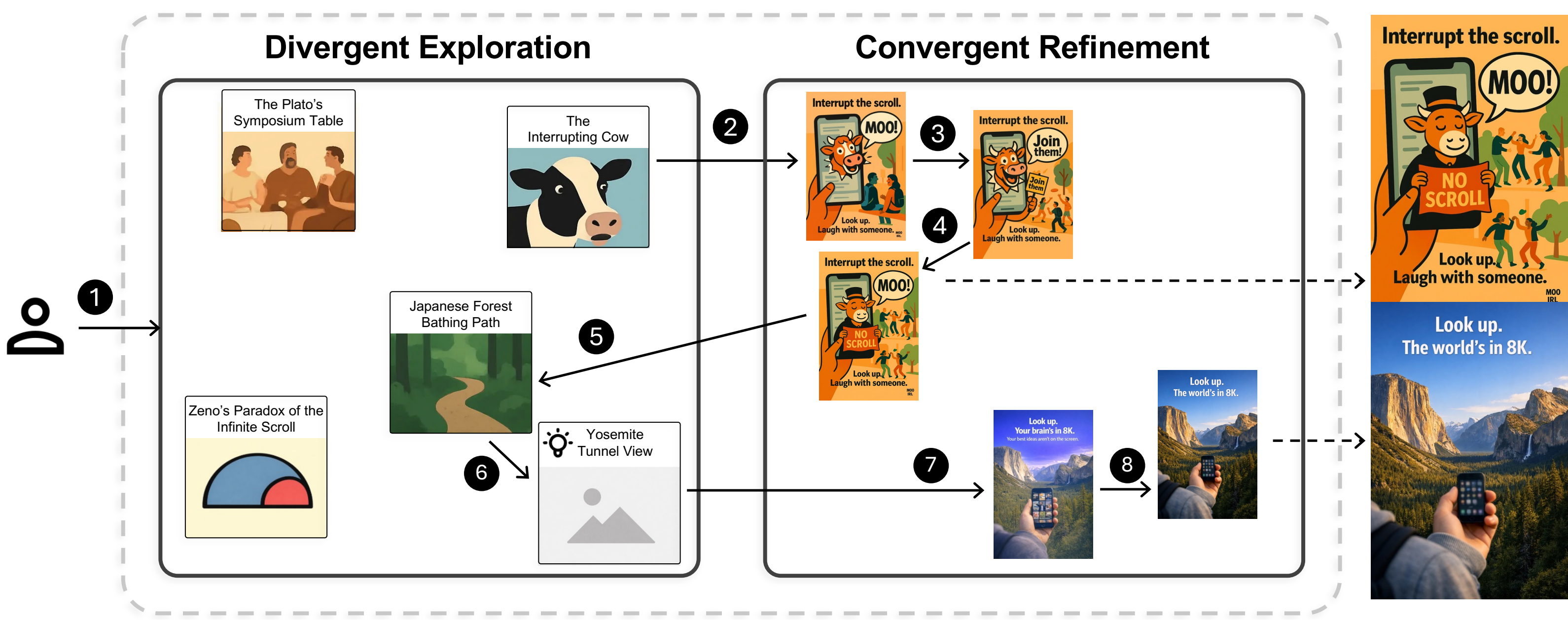
The Problem

- Chatbots can anchor users to early, less novel ideas.
- Exploration stays close to the initial task framing.
- Refinement assumes users know what they want.

Our Approach

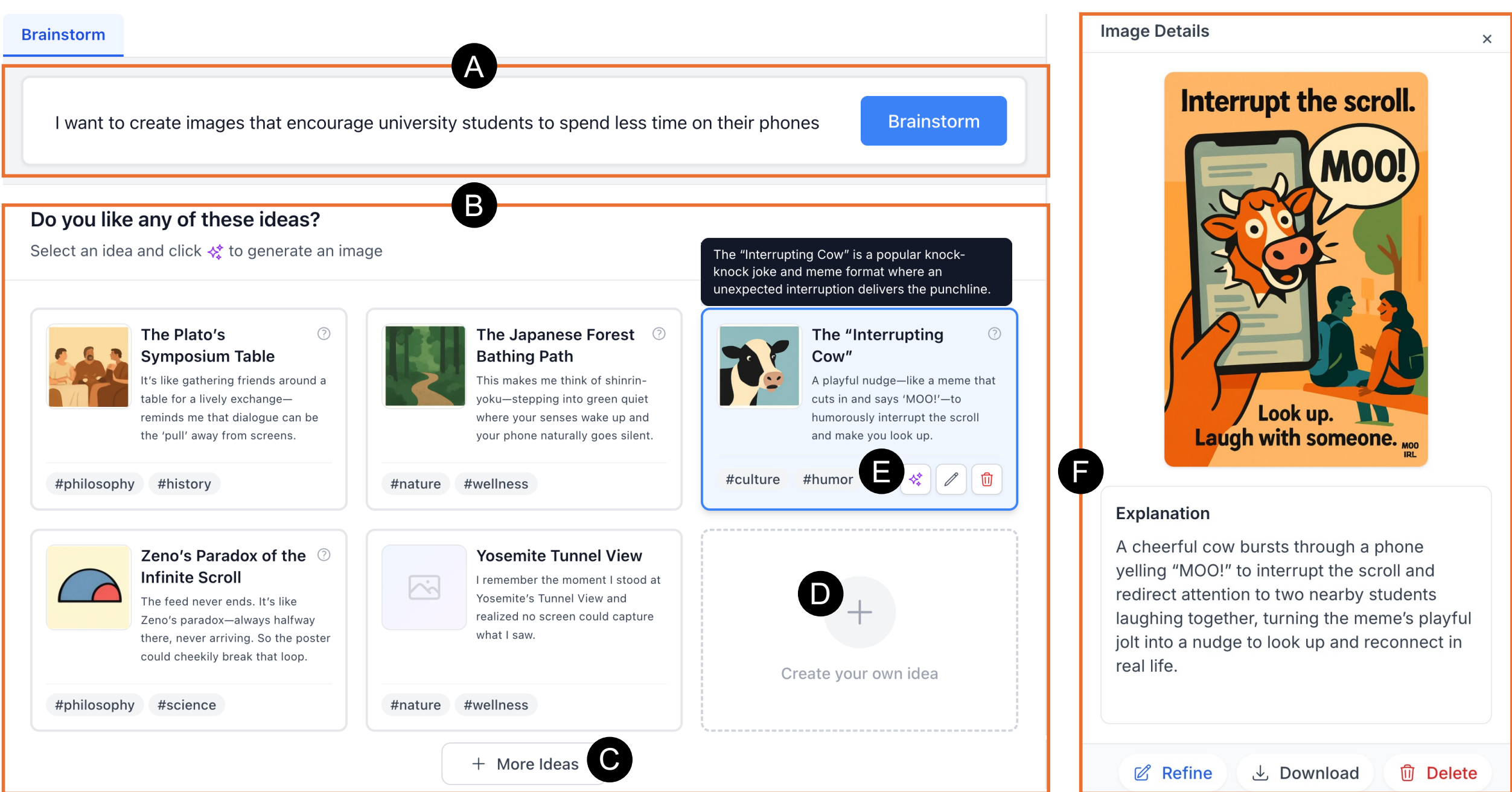
- HAICo structures interaction into two switchable modes: **DIVERGENT** and **CONVERGENT**.
- DIVERGENT**: support broad exploration by surfacing remote ideas by **associative thinking**.
- CONVERGENT**: support refinement via decomposing user intent into **structured options**.

HAICo Workflow

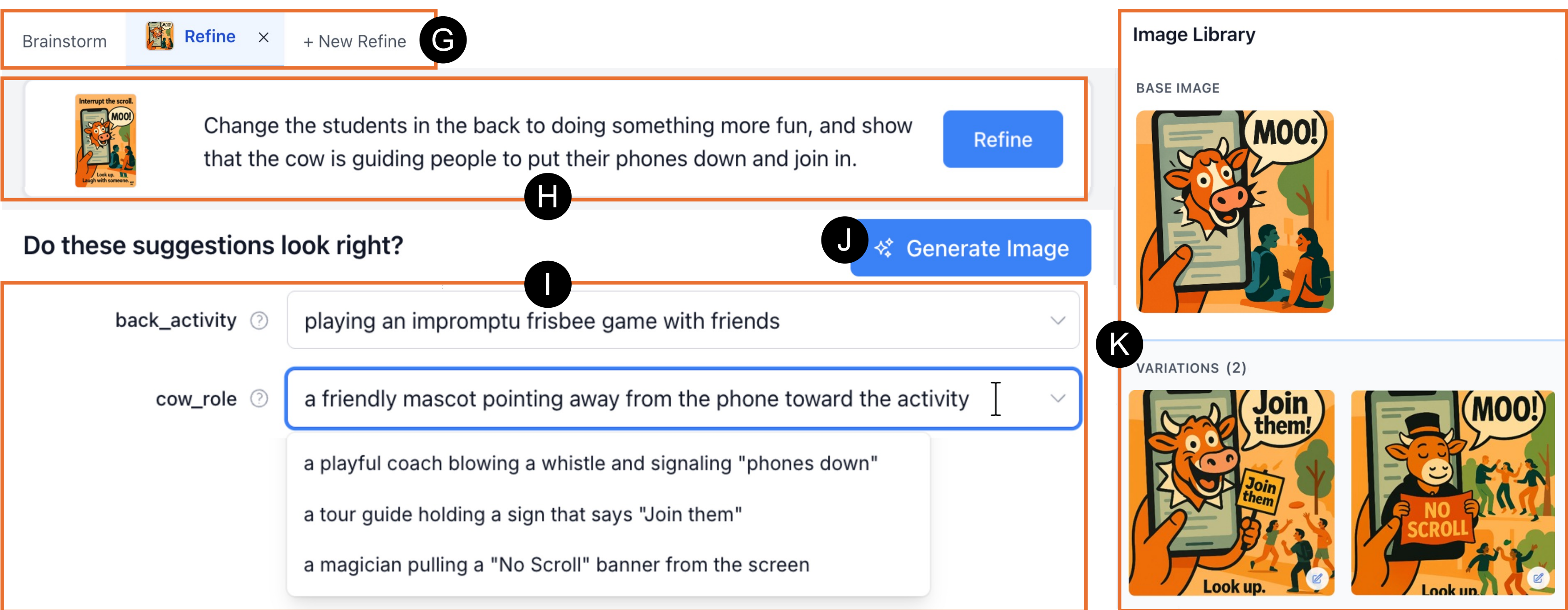


HAICo System: Two Switchable Modes

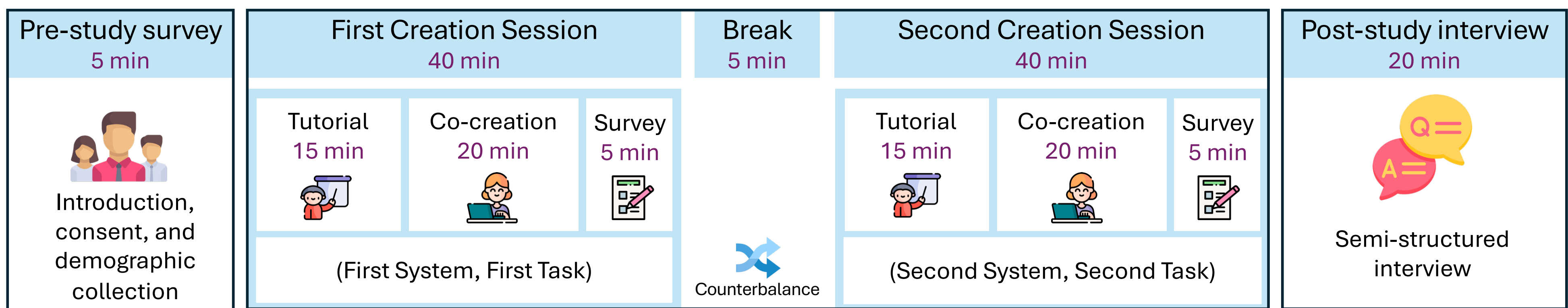
1 DIVERGENT: Expand the Exploration



2 CONVERGENT: Align with User Intent



User Study: Poster Design

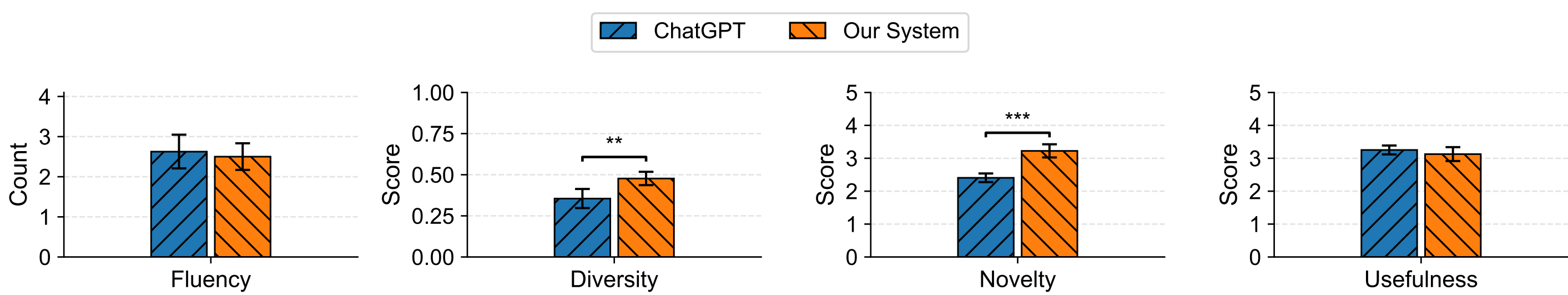


- Task:** design a poster promoting a behavior
- Study design:** within-subjects (HAICo vs. ChatGPT)
- Participants:** N = 24

Results: Creative Outputs

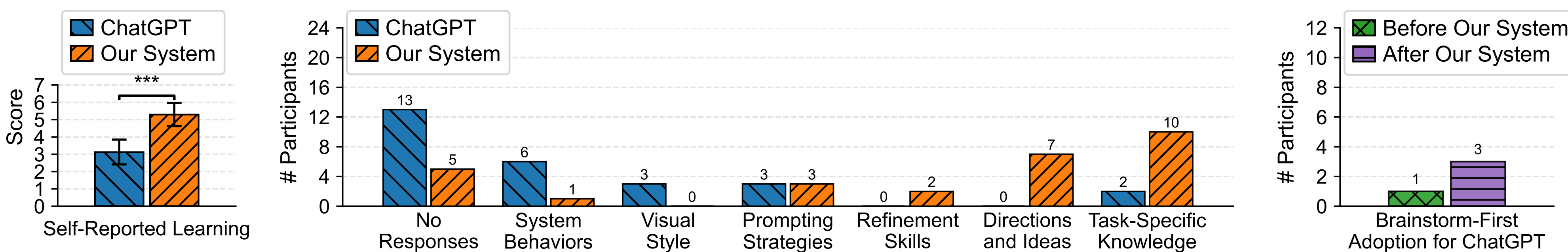


ChatGPT (best, median) HAICo (best, median)
Example posters created in the study for "spend less time on phones."



Images made with HAICo were significantly **more novel** (3.22 vs. 2.41, $p < .001$), and image sets were **more diverse** (0.48 vs. 0.36, $p = .001$); fluency and usefulness did not differ significantly.

Results: User Learning



Self-reported learning was higher with HAICo (5.29 vs. 3.12, $p < .001$) and of a *different kind*: ChatGPT users learned **about the system**, HAICo users **about the task**.

Discussion

- Human–AI Co-Exploration:** AI introduces unexpected directions; users adapt and build on them to expand the exploration space.
- Interactive Alignment:** Alignment is not matching a user prompt; user intent can be progressively specified through interaction.
- User Learning:** Co-creation systems should support both better outputs now and helping users become better creators over time.



Paper